

Project Milestone 2: Sales and Profit Analysis for Retail Decision Support Using the Global Superstore Dataset

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Revised Abstract

Retail organizations frequently rely on discount-based promotions to increase sales volume. However, increased sales do not necessarily translate into increased profitability. Excessive discounting may stimulate demand while simultaneously eroding margins. Therefore, data-driven approaches that prioritize profitability rather than revenue are essential for sustainable decision-making.

This project investigates transaction-level retail data from the Global Superstore dataset to identify key determinants of profit and evaluate predictive modeling strategies for decision support. The study integrates exploratory analysis with both regression and classification models. Regression is used to estimate continuous profit values, while classification models assess whether a transaction is likely to be profitable prior to execution. Model performance is compared against baseline benchmarks to ensure meaningful improvement.

By reframing retail analytics around profit risk rather than revenue growth, this study contributes a structured and evaluation-driven framework for pricing and discount decision support.

1. Literature Review

1.1 Retail Analytics and Predictive Decision-Making

The increasing availability of large transactional datasets has significantly transformed retail decision-making. Predictive analytics techniques are now widely applied in areas such as demand forecasting, pricing optimization, and customer segmentation (James et al., 2021; Géron, 2019). These approaches allow firms to move beyond descriptive reporting toward data-driven operational strategies.

However, much of the existing research prioritizes revenue growth and demand forecasting rather than direct modeling of profitability. While sales volume is an important performance indicator, it does not necessarily reflect financial sustainability. Profit, rather than revenue, ultimately determines long-term viability.

Wedel and Kannan (2016) argue that marketing analytics must evolve from descriptive dashboards toward predictive systems that actively support

managerial decision-making. In this context, models should not merely describe past performance but help anticipate future financial outcomes.

1.2 Promotional Strategies and Profitability

A substantial body of marketing research has examined the effects of promotional activities on sales performance. Blattberg, Briesch, and Fox (1995) demonstrate that while promotions can significantly increase short-term demand, their effect on profitability depends on cost structures, margin levels, and consumer responsiveness. Aggressive discounting may stimulate volume while compressing margins, potentially resulting in reduced overall profit.

Similarly, Gupta and Lehmann (2003) emphasize that firms should treat customers and transactions as long-term financial assets rather than focusing solely on short-term revenue gains. Their work reinforces the importance of evaluating financial impact beyond sales growth metrics.

Despite these findings, predictive modeling studies in retail analytics often prioritize forecasting quantity sold or total sales revenue. Fewer studies explicitly frame profitability as the primary outcome variable at the transaction level. This gap motivates the present study.

1.3 Predictive vs. Explanatory Modeling

In the broader analytics literature, an important distinction exists between explanatory and predictive modeling. Shmueli and Koppius (2011) argue that explanatory models aim to test theoretical relationships, whereas predictive models focus on generating accurate forecasts for future observations. In business contexts, predictive performance is often more relevant than causal interpretation, particularly when models are used for operational decision support.

This project adopts a predictive modeling framework. Rather than testing a causal hypothesis regarding discount effects, it evaluates how accurately profit and profitability can be predicted using observable transaction characteristics. Out-of-sample validation and baseline comparisons are therefore emphasized.

1.4 Regression and Classification in Business Analytics

Regression techniques are commonly applied to estimate continuous financial outcomes such as sales or profit (James et al., 2021). Linear regression provides interpretable relationships, while decision tree methods capture nonlinear patterns and interaction effects (Géron, 2019).

Classification algorithms, including logistic regression and decision trees, are frequently used to identify business risk scenarios such as churn, credit default, or fraud (Han et al., 2012; Raschka & Mirjalili, 2022). These methods categorize outcomes into predefined classes, allowing decision-makers to identify high-risk cases before adverse events occur.

Applying classification to transaction-level profitability reframes retail analytics as a risk detection problem. Instead of predicting fraud or churn, the objective becomes identifying transactions that are likely to generate financial loss.

1.5 Research Gap and Study Contribution

Although predictive modeling is well established in retail analytics, several limitations remain:

1. Revenue forecasting is more common than profit prediction.
2. Few studies combine continuous profit estimation with binary profitability classification.
3. Baseline benchmarking is not consistently emphasized in applied analytics exercises.

Furthermore, the Global Superstore dataset is frequently used for visualization or descriptive analysis but less often framed as a predictive decision-support case.

2. Data Description

2.1 Dataset Overview

The study uses the Global Superstore dataset obtained from Kaggle (Ilhan, n.d.).

Granularity:

Each observation represents a single product-level transaction within an order.

Time Span:

The dataset spans multiple consecutive years of retail sales activity.

Size:

The dataset contains 51,290 transaction records and 27 variables.

2.2 Variable Structure

Numerical Variables:

- Sales
- Profit
- Quantity
- Discount

Categorical Variables:

- Category
- Sub-Category
- Region
- Segment

2.3 Target Definitions

Two predictive targets are defined:

Regression Target:

Profit (continuous variable).

Classification Target:

Profitability label defined as:

- Profitable = $\text{Profit} > 0$
- Not Profitable = $\text{Profit} \leq 0$

This rule reflects a clear managerial interpretation of financial performance at the transaction level and ensures reproducibility in classification modeling.

2.4 Descriptive Characteristics

Initial exploratory analysis indicates:

- The dataset contains **51,290 transaction-level records and 27 variables**.
- Profit values include both positive and negative outcomes.
- Approximately **74.24% of transactions are profitable**, while **25.76% result in financial loss**.
- Profit distribution is skewed with notable outliers.
- Discount levels vary across product categories.
- Minimal missing values are present.

The presence of both profitable and non-profitable transactions supports the formulation of a binary classification task. Although the dataset is not severely imbalanced, the minority class (loss-making transactions) represents over one-quarter of observations. Therefore, evaluation metrics such as F1-score will be considered alongside accuracy.

2.5 Dataset Constraints

Several limitations must be acknowledged:

- The dataset does not include marketing expenditure data.
- Customer demographic information is unavailable.
- The data is observational; therefore, causal inference cannot be established.
- External economic conditions are not captured.

These constraints will be considered when interpreting modeling results.

3. Project Approach

3.1 Methodological Framework

The project will proceed through the following stages:

Stage 1: Data Preparation

- Data cleaning and validation
- Handling missing values
- Encoding categorical variables

- Feature selection

Stage 2: Exploratory Data Analysis

- Profit distribution analysis
- Discount–profit relationship assessment
- Category-level profitability comparison
- Correlation analysis

Stage 3: Predictive Modeling

Regression Models:

- Baseline: Mean profit predictor
- Linear regression
- Decision tree regression

Evaluation Metrics:

- Root Mean Squared Error (RMSE)
- Coefficient of determination (R^2)

Classification Models:

- Baseline: Majority-class predictor
- Logistic regression
- Decision tree classifier

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-score

Data Splitting Strategy:

- 80/20 train–test split

- Stratified sampling for classification
- Fixed random seed for reproducibility

3.2 Justification of Methodology

Regression modeling is appropriate for estimating continuous profit outcomes (James et al., 2021). Classification modeling supports the identification of high-risk transactions, enabling pre-emptive decision-making. Baseline comparisons ensure that predictive performance reflects genuine improvements beyond trivial heuristics. Tree-based models are included due to their ability to capture nonlinear relationships common in retail pricing data (Géron, 2019).

4. Visual Representation of Methodology

Data Collection



Data Cleaning



Exploratory Data Analysis



Feature Engineering



Regression Modeling



Classification Modeling



Model Evaluation



Managerial Insights

5. GitHub Repository

GitHub Repository Link:

<https://github.com/sahanashaikh14/global-superstore-profit-analysis>

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